

SESSION_07

Q.1 Use nested for loops to print a grid of emojis representing a 5x5 Instagram post feed, where each cell shows a 🐱 symbol.

A.1 #include <stdio.h>

```
int main() {
    int i, j;
    for (i = 1; i <= 5; i++) {
        for (j = 1; j <= 5; j++) {

printf(" 🐱 ");
        }
        printf("\n");
    }
    return 0;
}
```

Q.2 Build a right-angled triangle pattern using nested loops, where each row displays increasing numbers starting from 1, similar to how a leaderboard on a gaming app shows rank numbers.

A.2 #include <stdio.h>

```
int main() {
    int i, j;

    for (i = 1; i <= 5; i++) {
        for (j = 1; j <= i; j++)
        {
printf("%d ", j);
        }
        printf("\n");
    }
    return 0;
}
```

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Q.3 Create a pattern that prints a pyramid of stars (*) with 6 rows, centered like the loading animation you see on BookMyShow when a page is loading. Use spaces to align the stars in the center for each row.

```
A.3 #include <stdio.h>

int main() {
    int i, j, spaces;
    for (i = 1; i <= 6; i++) {

        // Print spaces
        for (spaces = 1; spaces <= 6 - i; spaces++) {
            printf(" ");
        }

        // Print stars
        for (j = 1; j <= 2 * i - 1; j++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

Alternating 0 and 1

```
#include <stdio.h>

int main() {
    int i, j;
    for (i = 1; i <= 4; i++) {
        for (j = 1; j <= 4; j++) {
            if ((i + j) % 2 == 0) {
                printf("0 ");
            }
            else {
                printf("1 ");
            }
        }
        printf("\n");
    }
}
```

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```
return 0;  
}
```

Q.5 Modify your pyramid pattern code to accept the number of rows as user input, so the user can set the height of the pyramid before printing.

A.5 #include <stdio.h>

```
int main() {  
    int rows, i, j, spaces;  
  
    printf("Enter the number of rows: ");  
    scanf("%d", &rows);  
  
    for (i = 1; i <= rows; i++) {  
  
        for (spaces = 1; spaces <= rows - i; spaces++) {  
            printf(" ");  
        }  
        for (j = 1; j <= 2 * i - 1; j++) {  
            printf("*");  
        }  
        printf("\n");  
    }  
  
    return 0;  
}
```
